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Introduction to Statistical Methods

ISM Team

QUESTION 1

- An autonomous vehicle company uses Simple Exponential Smoothing to forecast weekly data transmission volume (in TB) from its fleet of self-driving cars to cloud analytics servers. The transmission volume for the past six weeks is:

Week	1	2	3	4	5	6
Demand	112	118	125	121	130	142

The initial forecast for Week 1 equals the actual Week 1 demand.

Answer the following:

i. Forecast the demand for Week 7 using a smoothing constant $\alpha = 0.3$.

ii. Forecast the demand for Week 7 using a smoothing constant $\alpha = 0.8$.

iii. Briefly comment on which α value should be chosen when unexpected weather events cause sudden surges in sensor data uploads.

SOLUTION

Simple Exponential Smoothing - Solution

Given Data:

Actual Demand (TB) for Weeks 1 to 6: 112, 118, 125, 121, 130, 142

Initial Forecast F1 = A1 = 112

Case 1: $\alpha = 0.3$

Week	Actual Demand (A_t)	Forecast (F_t)
1	112	112.0
2	118	112.0
3	125	113.8
4	121	117.16
5	130	118.312
6	142	121.818
7	-	127.873

Forecast for Week 7 ≈ 127.87 TB

Case 2: $\alpha = 0.8$

Week	Actual Demand (A_t)	Forecast (F_t)
1	112	112.0
2	118	112.0
3	125	116.8
4	121	123.36
5	130	121.472
6	142	128.294
7	-	139.259

Forecast for Week 7 ≈ 139.26 TB

Conclusion

For sudden surges in demand, $\alpha = 0.8$ is preferred because it gives more weight to recent observations and responds quickly to changes.

Work Integrated Learning Programmes

QUESTION 2

A campus wellness app logs daily step counts (in thousands) of 10 students over a representative day:

{3.2, 3.8, 4.1, 4.5, 4.9, 9.2, 9.8, 10.3, 10.7, 11.4}. The data exhibits bimodality and is modeled using a two-component Gaussian Mixture Model (GMM).

Assume:

Component 1: First 5 values (sedentary students)

Component 2: Last 5 values (active students)

(a) Write the probability density function of a two-component GMM.

(b) Estimate the mean and variance for each component.

(c) Compute the mixing coefficients.

(d) Briefly explain how GMM can segment students for targeted health intervention programs.

Gaussian Mixture Model (GMM) – Campus Wellness App Case Study

(a) Probability Density Function of Two-Component GMM

A two-component Gaussian Mixture Model (GMM) is defined as:

$$p(x) = \pi_1 N(x \mid \mu_1, \sigma_1^2) + \pi_2 N(x \mid \mu_2, \sigma_2^2)$$

Where:

π_1, π_2 = Mixing coefficients ($\pi_1 + \pi_2 = 1$)

μ_1, μ_2 = Means of the components

σ_1^2, σ_2^2 = Variances of the components

$$N(x \mid \mu, \sigma^2) = (1 / \sqrt{(2\pi\sigma^2)}) * \exp(-(x - \mu)^2 / (2\sigma^2))$$

(b) Mean and Variance Estimation

Component 1 (Sedentary Students): {3.2, 3.8, 4.1, 4.5, 4.9}

Mean (μ_1) = 4.1, Variance (σ_1^2) = 0.34

Component 2 (Active Students): {9.2, 9.8, 10.3, 10.7, 11.4}

Mean (μ_2) = 10.28

Variance (σ_2^2) ≈ 0.566

(c) Mixing Coefficients

Total students = 10

Students in Component 1 = 5

Students in Component 2 = 5

$\pi_1 = 5/10 = 0.5$

$\pi_2 = 5/10 = 0.5$

(d) Role of GMM in Student Segmentation

The Gaussian Mixture Model segments students based on step count patterns.

- Component 1 represents sedentary students (average ≈ 4.1 thousand steps).
- Component 2 represents active students (average ≈ 10.28 thousand steps).

The wellness app can use this segmentation to:

- Identify low-activity students for targeted fitness programs.
- Provide personalized step goals.
- Design health intervention campaigns based on activity levels.

Thus, GMM enables probabilistic clustering and supports personalized health interventions.

QUESTION 3

The following time-series represents the residuals obtained after fitting a forecasting model:

{3, 2, 1, 0, −1, −2, −3, 0}

i. Compute the sample mean and the sample variance of the series.

ii. State whether the series can be considered white noise. Justify briefly.

SOLUTION

Time-Series Residual Analysis

Residual series: {3, 2, 1, 0, −1, −2, −3, 0}

Number of observations (n) = 8

i) Sample Mean and Sample Variance

Sample Mean:

$$\bar{x} = (3 + 2 + 1 + 0 - 1 - 2 - 3 + 0) / 8 = 0 / 8 = 0$$

Sample Variance:

$$s^2 = \Sigma(x_i - \bar{x})^2 / (n - 1)$$

Since $\bar{x} = 0$,

$$s^2 = (9 + 4 + 1 + 0 + 1 + 4 + 9 + 0) / 7 = 28 / 7 = 4$$

ii) White Noise Justification

A white noise series should have:

- Mean approximately equal to 0
- Constant variance
- No autocorrelation (random pattern)

In this case, the mean is 0 and variance is constant. However, the residuals show a clear decreasing pattern (3, 2, 1, 0, −1, −2, −3), which indicates dependence between observations.

Therefore, the series cannot be considered white noise because it shows a systematic trend instead of random fluctuations.